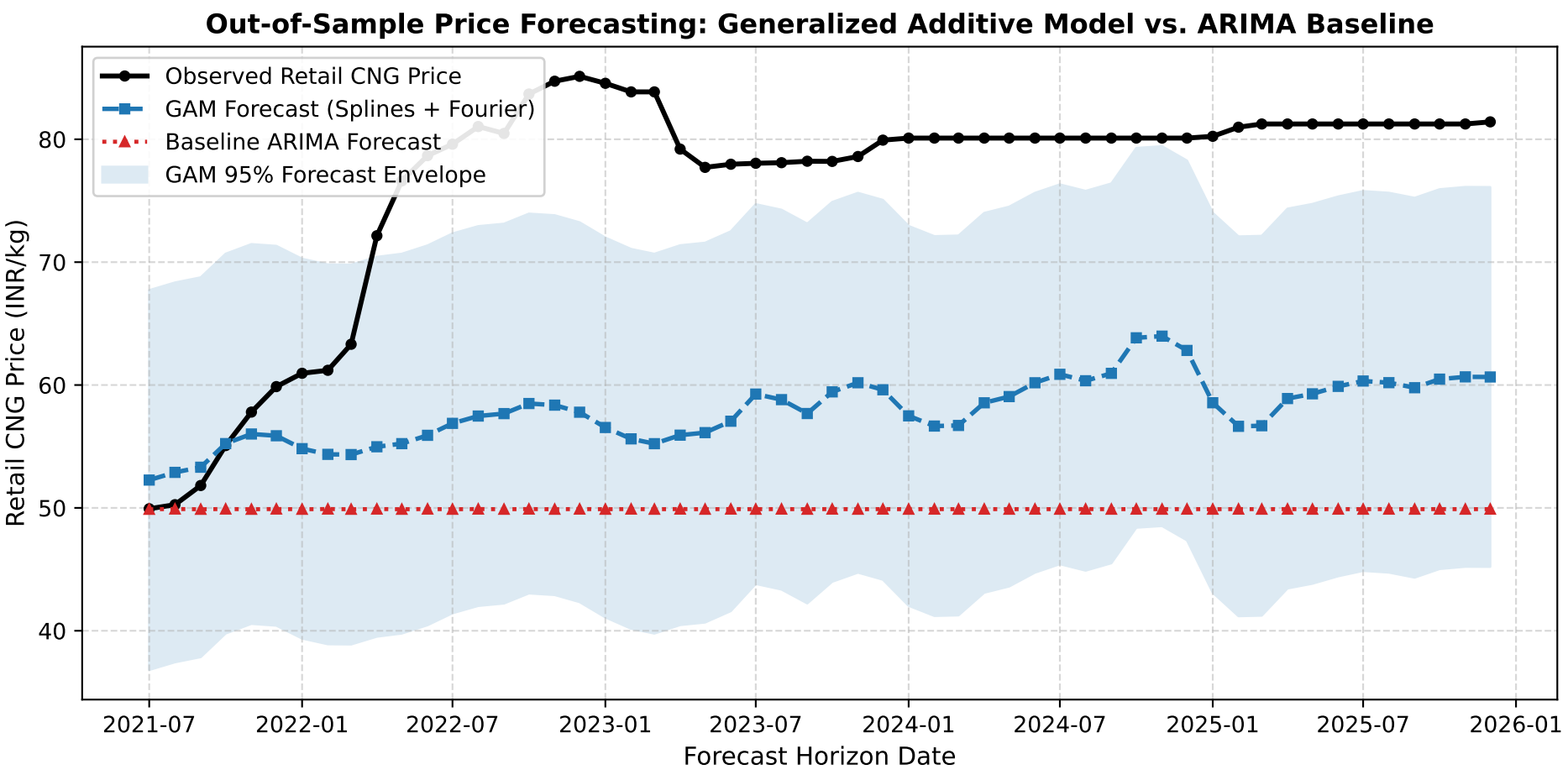
Out-of-Sample Price Forecasting: Generalized Additive Model vs. ARIMA Baseline
Observed Retail CNG Price
GAM Forecast (Splines + Fourier)
Baseline ARIMA Forecast
GAM 95% Forecast Envelope
Retail CNG Price (INR/kg)
80
70
60
50
40
2021-07
2022-01
2022-07
2023-01
2023-07
2024-01
2024-07
2025-01
2025-07
2026-01
Forecast Horizon Date